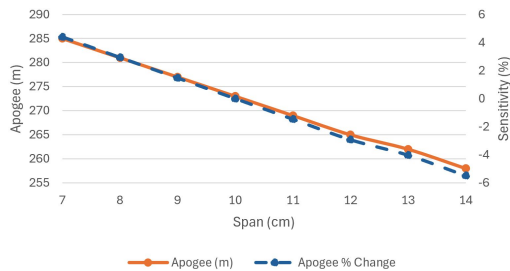


Aero-Performance Sensitivity Study

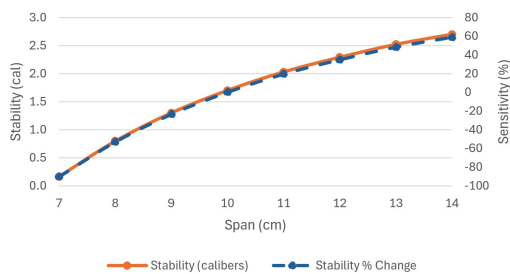
Span, Sweep, and Tip Chord

Span:

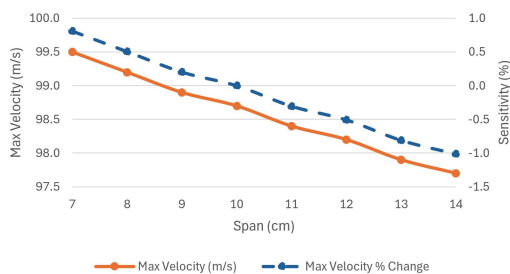
Span vs. Apogee & Sensitivity



Span vs. Stability & Sensitivity

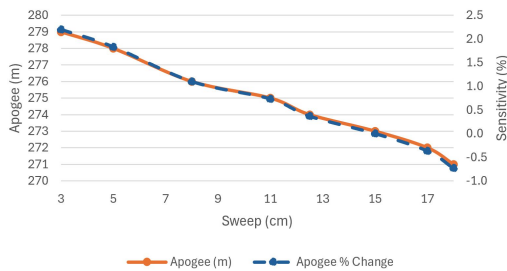


Span vs. Max Velocity & Sensitivity

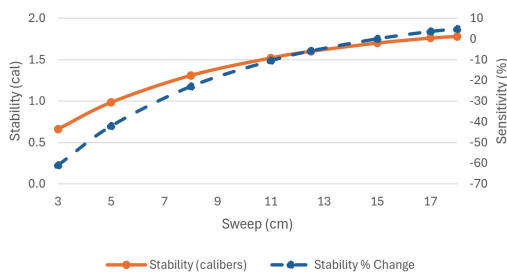


Sweep:

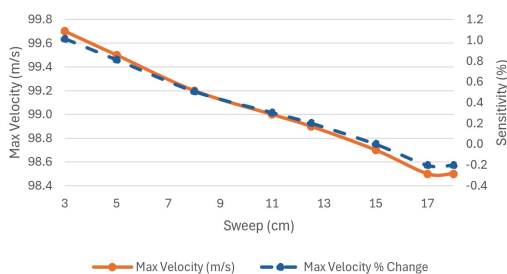
Sweep vs. Apogee & Sensitivity



Sweep vs. Stability & Sensitivity

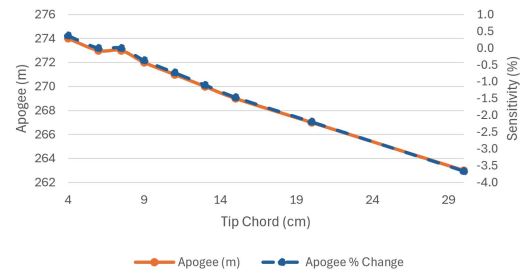


Sweep vs. Max Velocity & Sensitivity

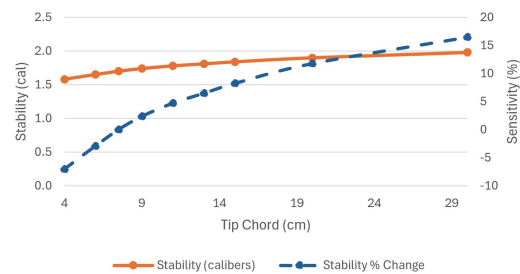


Tip Chord:

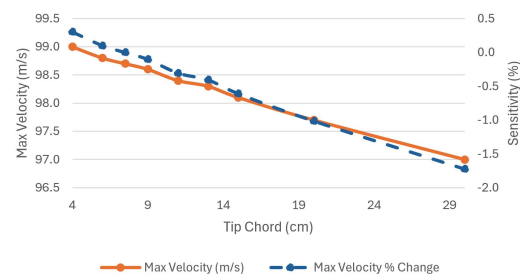
Tip Chord vs. Apogee & Sensitivity



Tip Chord vs. Stability & Sensitivity



Tip Chord vs. Max Velocity & Sensitivity



Fin span was tested from 7 to 14 cm with a baseline value of 10 cm. Increasing span steadily improves stability while reducing apogee and max velocity. Span exhibits the highest stability sensitivity of all parameters, dropping 90% when reduced from the baseline down to 7 cm.

Sweep was tested from 3 to 18 cm around a 15 cm baseline. Like span, increasing sweep improves stability while slightly decreasing the apogee and the max velocity. The sweep stability shows moderate sensitivity, dropping 61% when reduced from baseline down to 3 cm.

Tip chord was tested from 4 cm to 30 cm with a baseline value of 7.5 cm. Adjusting tip chord yields the lowest overall sensitivity across the entire study, making it the ideal parameter for minor geometry tweaks without significantly changing the flight performance.

Conclusion: Overall sensitivity analysis reveals that fin span dominates vehicle aerodynamics, driving a 90% variation in stability caliber from the baseline, while sweep and tip chord exhibit moderate (61%) and minimal (16%) stability impacts, respectively. Across all geometries, gains in stability caliber consistently introduce aerodynamic drag that slightly reduces peak apogee and maximum velocity. Therefore, an optimal fin design workflow dictates sizing tip chord first to lock in aerodynamic efficiency and structural stiffness against flutter, adjusting the sweep to mitigate high speed drag, and finally tuning the span to satisfy the primary stability constraints.